KAGGLE COMPETITION TECHNICAL REPORT

March Machine Learning Mania 2026

Predicting NCAA Basketball Tournament Outcomes Using a 5-Stage Machine Learning Pipeline

Key Results

0.03037	**87.9%**	**0.1698**
Stage 1 Brier Score	vs. Naive Baseline	Live Leaderboard Brier
0.8785	**78**	**0.321**
Stage 1 Skill Score	Features per Matchup	Live Brier Skill Score

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1 Introduction

The Kaggle *March Machine Learning Mania 2026* competition requires calibrated win-probability estimates for all 132,133 possible NCAA tournament matchups (66,430 men's + 65,703 women's), evaluated by Brier score ($= \frac{1}{N}\sum(p_i - y_i)^2$, lower is better; naive baseline = 0.25, prize pool $50,000). Two pipeline versions were developed iteratively. V1 established a baseline on competition data only; V2 redesigned every component and achieved the final submitted predictions. The live Kaggle leaderboard score on actual 2026 tournament games was **0.1698** (BSS = 0.321).

2 The Brier Score and Why Calibration Is Everything

The Murphy (1973) decomposition shapes every modelling decision in this work:

$$BS = \underbrace{\text{Reliability}}_{\text{calibration error}} - \underbrace{\text{Resolution}}_{\text{discrimination}} + \underbrace{\text{Uncertainty}}_{\text{irreducible}}$$

- **Uncertainty** is fixed, we cannot control how many upsets happen.
- **Resolution** measures how well we separate strong teams from weak ones.
- **Reliability** measures calibration. A model saying 0.95 when teams win 65% of the time at that confidence contributes a 49× larger penalty than a perfectly calibrated model. This is why calibration received equal engineering effort as feature construction.

The Brier Skill Score $BSS = 1 - BS\,/\,\bar{o}(1-\bar{o})$ normalises against the naive baseline; BSS = 1 is perfect, BSS = 0 is random guessing.

3 Data and Tools

External data sources:

- KenPom (paid): AdjEM, AdjO, AdjD, Luck, Tempo
- Barttorvik (free): Barthag, AdjO, AdjD, AdjT
- Vegas odds (manual): market win probabilities
- Massey Ordinals: median of 100+ rankings

Stack: Python 3.12 on Google Colab (free tier). Libraries: `xgboost`, `catboost`, `optuna`, `scikit-learn`, `scipy`, `kenpompy`, `cloudscraper`, `beautifulsoup4`, `pandas/numpy`, `matplotlib/seaborn`. A JavaScript keep-alive script simulated browser activity every 60 seconds to prevent Colab timeouts during the 30–40 minute runtime.

4 Pipeline V1 — Baseline (Stage 1 Brier: 0.0819)

V1 used competition data only: Elo ratings, season stats, seeds, and Massey ordinals as Team A − Team B feature differences. Five models (LR, XGBoost, LightGBM, CatBoost, Random Forest) were blended with inverse-squared Brier weights.

Stage 1 historical validation: Brier = 0.0819, BSS = 0.672. Despite solid CV numbers, V1 ranked near the bottom of the leaderboard. Root causes: (1) overconfident uncalibrated predictions inflating the Reliability term; (2) no external data; (3) five correlated models adding noise without improving Resolution on only ∼630 training samples.

5 Pipeline V2 — Redesigned Ensemble

5.1 Model Reduction: 5 → 3 Models

With 602 training samples, correlated models increase the Reliability penalty without improving Resolution. LightGBM (near-duplicate of XGBoost), Random Forest (weak on small tabular data),

Table 1: V1 cross-validation results — Men's (LOSO 8 folds, 2018–2025)

Model	Mean Brier	Std	vs Baseline
Logistic Regression	0.2042	±0.0272	+18.3%
XGBoost	0.2103	±0.0331	+15.9%
LightGBM	0.2177	±0.0337	+12.9%
CatBoost	0.2051	±0.0282	+18.0%
Random Forest	0.1998	±0.0227	+20.1%
Ensemble	∼0.197	—	+21%
Naive (0.5)	0.2500	—	—

and any MLP (far too few samples) were removed. Meta-weights are Optuna-learned via stacking CV:

Model	Meta-weight	Best CV Brier	
XGBoost	59.2%	0.0977	(depth=4, lr=0.041, n_est=838)
CatBoost	35.6%	0.0976	(depth=6, lr=0.030, iters=805)
Logistic Regression	5.1%		calibration anchor

Hyperparameters tuned with Optuna TPE (60 XGB + 50 CatBoost = 110 trials), minimising leave-one-season-out Brier on the 4 most recent training seasons.

5.2 Feature Engineering — 78 Features, 11 Groups

Table 2: Feature groups in V2 (78 features total per matchup)

Feature Group	Count	Top Feature (Pearson r)
Elo (current + trend)	4	`diff_Elo`: 0.643
Season stats (time-decay)	18	`diff_WinPct_wtd`: 0.313
Advanced box score	14	`diff_TrueShooting`: 0.179
Tempo-free ratings	9	`diff_NetRtg`: 0.388
Bradley-Terry	4	`bt_logratio`: 0.453
Seeds	5	`diff_Seed`: 0.459
Massey Ordinals	2	`diff_Massey`: 0.413
KenPom	6	`kp_em_diff`: 0.377
Barttorvik	5	`tv_net_diff`: 0.341
Vegas odds	3	`vegas_logit_diff`: 0.298
Conference + Coaches	8	`diff_ConfStrength`: 0.375
Total	**78**	`elo_win_prob` = **0.669 (highest)**

Key engineering decisions: Elo enhanced with margin-of-victory multiplier ($K_{\text{eff}} = K(1 + 0.04\Delta)$, capped at $3K$), home-court adjustment (+100 Elo in expected score), and 30% mean reversion between seasons. Tempo-free stats use the Dean Oliver possession estimate ($\widehat{\text{Poss}} \approx \text{FGA} - \text{OReb} + \text{TO} + 0.44 \times \text{FTA}$). Time-decay weights each game by $w = (\text{DayNum}/132)^{1.5}$ (March games $\approx 4\times$ early-season weight). Season averages use a 4:2:1 exponential decay over three prior seasons.

5.3 Stacking and Anti-Leakage CV

Why not random K-fold? Features for season S use seasons $S-1$, $S-2$, $S-3$. Random splits mix same-cohort games into train and test, inflating CV scores and corrupting model selection. Leave-one-season-out (LOSO) CV is the only correct approach: for each validation season s, all three base models train on all other seasons and generate out-of-fold (OOF) predictions; a meta-learner (LR, $C = 1.0$) trains on the stacked OOF matrix with zero leakage. StandardScaler is fitted on the training fold only.

5.4 Three-Method Calibration

Three methods competed on OOF Brier; **isotonic regression won for both genders**. Beta calibration fits $\text{logit}(p_{\text{cal}}) = a \log p - b \log(1-p) + c$ by MLE. Bayesian Beta-prior smoothing uses the posterior mean $p_{\text{cal}} = (\alpha + p_{\text{raw}} \cdot n_e)/(\alpha + \beta + n_e)$, shrinking extreme predictions toward the base rate; (α, β, n_e) minimise OOF Brier directly.

6 Results

Table 3: V2 stacking CV results — Men's and Women's

Model	Men's Brier	Men's vs Base	Women's Brier	Women's vs Base
Logistic Regression	0.1108	+55.7%	0.0813	+67.5%
XGBoost	0.1025	+59.0%	0.0802	+67.9%
CatBoost	0.1071	+57.1%	0.0767	+69.3%
Meta-learner (raw)	0.0999	+60.1%	0.0744	+70.2%
After isotonic	**0.0918**	**+63.3%**	**0.0676**	**+72.9%**
Naive baseline	0.2500	—	0.2500	—

Table 4: Version comparison — Stage 1 historical validation (536 games, 2021–2025)

Pipeline	Stage 1 Brier	BSS	vs Baseline
Naive (always 0.5)	0.2500	0.000	—
V1 (5 models, no external data)	0.0819	0.672	+67.2%
V2 (3 models + KenPom + isotonic cal.)	**0.0304**	**0.879**	**+87.9%**
Perfect	0.0000	1.000	+100%

Table 5: Top 15 features by XGBoost importance and Pearson correlation (Men's)

Rank	Feature	Importance	Correlation
1	`elo_win_prob`	0.118	0.669
2	`diff_Elo`	0.097	0.643
3	`diff_EloWinProb`	0.088	0.620
4	`diff_Seed`	0.079	0.459
5	`diff_BT_log`	0.071	0.453
6	`bt_logratio`	0.068	0.453
7	`diff_Massey`	0.062	0.413
8	`diff_Elo_prev`	0.057	0.392
9	`diff_NetRtg`	0.051	0.388
10	`kp_em_diff`	0.048	0.377
11	`diff_ConfStrength`	0.043	0.375
12	`diff_KP_AdjEM`	0.041	0.376
13	`seed_A`	0.038	0.368
14	`diff_KP_AdjO`	0.035	0.359
15	`diff_AvgPtsDiff`	0.032	0.342

7 Submission, Leaderboard, and Discussion

`submission_final_combined_2026.csv` (132,133 matchups, Stage 1 + Stage 2) was submitted before the March 19, 2026 deadline. All predictions were clipped to $[0.03, 0.97]$.

Metric	Value	Notes
Stage 1 Brier (historical 2021–2025)	0.03037	536 matched games
Stage 1 Brier Skill Score	0.8785	+87.9% vs baseline
Live Leaderboard Brier (2026)	**0.1698**	actual 2026 tournament
Live Brier Skill Score	**0.321**	32% above random
Features per matchup	78	across 11 groups
Training samples (men's)	602	2016–2025
Total 2026 predictions	132,133	both genders

The gap between Stage 1 (0.03037) and the live score (0.1698) is expected. Stage 1 evaluated on years partly overlapping with the calibration and HPO optimisation window, an inherently optimistic estimate. Live scoring is a true out-of-distribution test: 2026 features relied on a single completed regular season and the model never saw those games during training. A live BSS of 0.321 is a real, honest result on games nobody had seen.

8 Summary and Conclusions

Key Lessons Learned

1. **Calibration dominates on small datasets.** Fixing the Reliability term improved Brier more than any single model change.
2. **External data beats model complexity.** KenPom alone contributed ∼15–20% — more than adding four extra models.
3. **Fewer, diverse models win with 602 samples.** 3 models > 5 correlated models on this data.
4. **LOSO CV is non-negotiable.** Random K-fold creates feature leakage and corrupts model selection.
5. **Feature engineering beats tuning.** Time-decay, Bradley-Terry, conference strength, and coach continuity added signal no Optuna trials could manufacture.

Final Results Summary

Stage 1 Brier Score (historical 2021–2025)	**0.03037**
Stage 1 Brier Skill Score	**0.8785**
Improvement vs. baseline	**+87.9%**
Live Leaderboard Brier (2026)	**0.1698**
Live Brier Skill Score	**0.321**
Features per matchup	**78**
Training samples (men's)	**602**
Total 2026 predictions	**132,133**

All methodology, results, and analysis are original work.